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# The impact of near-real-time deforestation alerts across the tropics

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**Reducing deforestation to mitigate climate change necessitates monitoring of deforestation activity. However, while freely available deforestation alerts on forest loss are available, the effect of these alerts and the presence of subscribers in a particular area is unclear. Here, we show that subscriptions to alerts in 22 tropical countries decrease the probability of deforestation in Africa by 18% relative to the average 2011–2016 levels. There is no effect on other continents, and the availability of alerts does not significantly change deforestation outcomes. This decrease in Africa is higher in protected areas and concessions, suggesting that alerts either increased capacity to enforce existing deforestation policy or induced the development of more effective anti-deforestation policies. Calculated using the social cost of carbon for avoided deforestation in Africa, we estimate the alert system's value to be between US$149 million and US$696 million.**

Cost-effective reduction of deforestation should be a cornerstone in the suite of climate change mitigation strategies. Land-use change accounts for 6–17% of global carbon emissions[1]. At any reasonable carbon price, avoided deforestation provides 7.2–9.6 times the abatement potential of reforestation[2]. There are a range of strategies for accomplishing this—protected areas[3], payments for ecosystem services[4] and supply-chain initiatives[5], among others. However, all of these strategies require monitoring of deforestation activities. Given that deforestation rates are often highest in low- and middle-income countries with limited resources to create effective monitoring systems[6], the availability of low-cost monitoring technologies may be an important support to policy aimed at avoiding deforestation.

This Analysis estimates the impact of the Global Land Analysis and Discovery (GLAD) alerts of tree cover loss[7] on deforestation trends across 22 tropical countries between 2011 and 2018. The system that we examine is the first to offer loss alerts at high spatial resolution (~30 m) and frequency (up to every 8 days, depending on cloud cover). The GLAD alerts use all available Earth observation data from Landsat 7 and Landsat 8 satellites, which combined have a revisit rate of 8 days. Each day, all new images are first masked for clouds and shadow, and then a loss detection algorithm is applied to all unobscured land, evaluating the reflectance of the current observation together with baseline Landsat metrics. Each 0.00025° pixel (28 m at the Equator) flagged as loss becomes an alert. The alerts map all forms of tree cover loss including logging, clear-cuts and intensive fires in both natural forests and plantations. These alerts and the date of first detection are provided without charge through a simple interface called Global Forest Watch (GFW).

Available in Chinese, English, French, Indonesian, Portuguese and Spanish, GFW also enables users to select a specific area for which they can receive alerts via e-mail. The e-mails state the number of pixels with suspected deforestation within the subscribed area, list the ten most recent alerts with geographic coordinates and include a hyperlink to view the locations on a map, as well as a link to download a file containing all alerts with coordinates and detection dates (Supplementary Fig. B1). Access to and use of the alerts require relatively low technical capacity. They can be downloaded on a cellphone during a temporary Internet connection and a cellphone GPS without Internet can be used to navigate to the site of probable deforestation. Before GLAD alerts, most ongoing prevention of deforestation relied on voluntary reports and/or ranger patrols. Because rangers and volunteers are not omnipresent, this undermined policymakers' ability to locate emerging deforestation hotspots.

Assuming that free deforestation alerts reduce the cost to policymakers of monitoring forests, thereby reducing the cost of implementing deforestation policy, we examine two questions: whether making these alerts available affected deforestation rates, and whether areas that were actively monitored by subscribers using the system saw a change in deforestation trends. Recent work[8] in Brazil has shown the substantial contributions of the government's own real-time monitoring system to the reduction in national deforestation trends. However, it is not known whether providing deforestation monitoring information freely via a platform unaffiliated with any governmental institution and accessible from anywhere on the globe could have an impact on land-use trends. Here we show that under certain conditions, it can.

We use a random sample of 1 km$^2$ grid cells from all countries that began receiving alerts before 2018, and estimate the impact of GLAD alerts on deforestation using two different strategies. We first examine whether having access to GLAD alerts for a particular region decreases deforestation using the gradual rollout of the alert system that began in 2016 (see timeline in Fig. 1a). Second, we study whether using GLAD alerts via subscription results in a decrease in deforestation rates. In the case of both access and use, we compare trends in areas covered earlier with those covered later, holding constant factors known to affect deforestation, such as location and biophysical characteristics. Figure 1b shows the countries analysed for this study and the distribution of forest within them. Our main outcome is the annual probability of forest loss[9] extracted at the grid-cell level for 2011 through 2018

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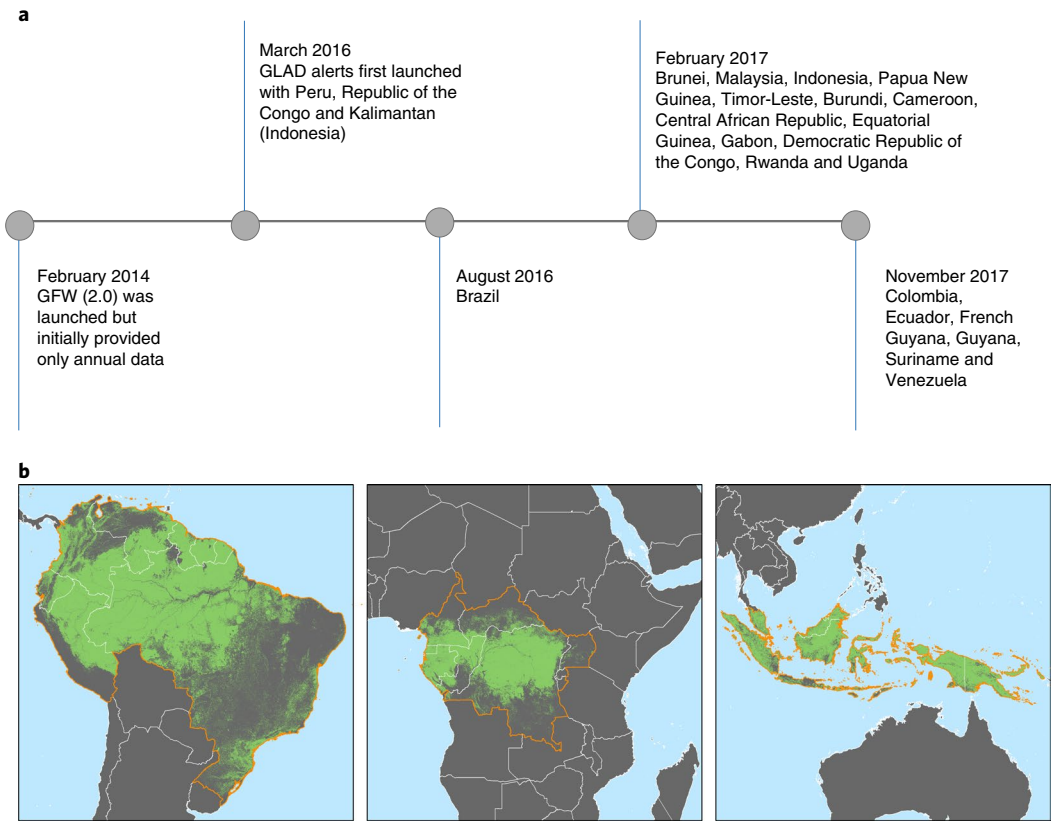



**Fig. 1 | Timeline of GLAD rollout and study region. a**, The rollout of GLAD alerts on GFW. **b**, The forest cover within our study region in 2010 (green shading). Study countries are outlined in orange; we exclude the Amazon biome of Brazil in our estimations.

(this outcome does not distinguish between natural and anthropogenic sources of forest loss). This measure is equal to one if there is a positive amount of deforestation during a given year, zero if there is none.

To evaluate the impact of alerts, we examine the data in two ways. First, we use the country as the unit of analysis and apply country fixed effects and year-by-continent fixed effects. Second, we use the grid cell as the unit of analysis and parallel the country model. We apply grid-cell and year-by-continent fixed effects, annual measures of average temperature, cumulative precipitation and the same flexible time trends, with covariates measured at the grid-cell level. Our approach eliminates the effect of time-invariant characteristics that influence deforestation (for example soil suitability, slope and so on), general macroeconomic events affecting all countries or specific regions, impacts of weather and confounding variation resulting from changes in transport or production technologies over time. More details can be found in the Methods.

## Alert availability does not affect deforestation rates

To estimate the impact of alert availability, we examine only countries that had access to alerts by September 2018. In this case, the counterfactual is countries that received alerts later. If it were the case that forests with higher deforestation rates received alerts sooner than forests with lower rates, and if those rates continued to be the same over time, then even in the absence of an effect of alerts, the estimated impact (approximately the deforestation rate of early recipients minus that of late recipients) would show that the availability of alerts had increased deforestation. In this case, the comparison we make would understate the impact of alert availability. The reverse is true if forests at low risk of deforestation receive alerts first.

Tests of differences in deforestation rates between late- and early-enrolled countries before 2016 suggest that the former provide a plausible counterfactual for the latter, as there are no significant differences in deforestation in this pre-alert period (Supplementary Tables A3–A5). If we assume that time trends continue in a similar way across comparison groups in the absence of the alerts, then our results are not likely to be biased.

There is no robust evidence of any impact of alert availability on average or in any particular region (Fig. 2 and Supplementary Table A2). Results using the alternative outcomes of per cent forest loss and winsorized per cent forest loss are qualitatively similar to the binary outcome estimates (Supplementary Tables A6 and A7). As it is plausible that countries may take time to adapt to the availability of alerts, we also examine a specification with a 1 yr lag of alert availability (Supplementary Tables A8–A10). Although results from this specification are not statistically different from zero, the sign of the average effect switches to become negative.

## Lower deforestation in African forests with subscriptions

To assess the impact of using GLAD alerts for monitoring, we first identified users who have the ability to act on alert information. These users, through their subscriptions to GFW, can select either an existing jurisdiction, upload a shapefile or identify the boundaries of a region about which they wish to receive alerts. Leveraging answers to voluntary questions, we divide subscriptions into two categories: those with intent to control deforestation and those without. Those without intent include students, academics or staff from the World Resources Institute (WRI) or affiliates, whereas those with intent to control include all other subscriptions to GLAD attached to areas smaller than 100 Mha. The spatial distribution and total number of subscriptions with and without



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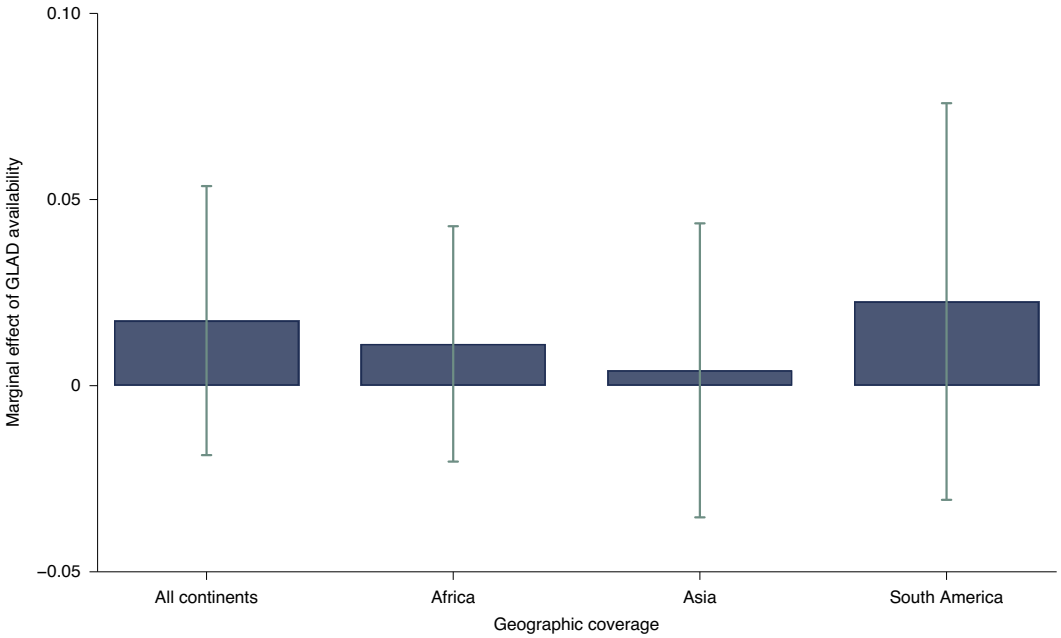



**Fig. 2 | Impact of GLAD availability on deforestation.** The point estimates (shaded bars) and 95% confidence intervals are shown for the impact of GLAD availability on the probability of deforestation on average for all continents are shown as well as for Africa, Asia and South America. Each bar shows the impact from a different estimation. The first is from column 6 in panel A of Supplementary Table A2 and the other three are from column 6 in Panel B of the same table, showing the total effect (that is, the linear combination of the baseline and interaction effects).

intent are presented in Fig. 3. We provide summary statistics and discussion of these different classifications in the Methods and Supplementary Section B1.

We restrict the sample to the subgroup of forested grid cells from areas that have GLAD subscriptions with intent to control by 2018 (the results of those without intent are in Supplementary Tables B8–B11). In contrast to the previous estimation, the treatment variable in this analysis varies within countries. Because the sample is composed only of areas that ever had a GLAD subscription, we eliminate one source of bias: the situation where forests at higher risk of deforestation are more likely to be monitored than forests at lower risk of deforestation. However, the approach does not eliminate the possibility that forests at higher risk of deforestation may be monitored sooner than forests at lower risk of deforestation. If this is the case, using the same logic described above, our estimates provide a lower bound of the effect of subscription to the GLAD alerts on deforestation.

Summary statistics suggest that areas with early subscriptions tend to have had slightly higher deforestation before alert availability (Table 1 and Supplementary Section B2). Pre-alert trends across the different subscription groups (Supplementary Tables B3–B5) were not statistically significantly different using the binary deforestation and winsorized outcomes, but show significant differences for the per cent deforestation outcome. This supports the validity of comparisons for the first two outcomes but not for per cent deforestation. Additional robustness checks detailed in the Supplementary Information confirm large, negative and statistically significant effects on forest in Africa.

The average effect of subscriptions on deforestation is negative (the opposite sign of the impact of alert availability), but statistically insignificant and small compared with the average yearly 2011–2016 deforestation probability (0.18). Results using winsorized per cent deforestation and per cent deforestation as outcomes tell the same story (Supplementary Tables B6 and B7). The effect of subscriptions on deforestation is probably a conservative estimate because some areas are likely to receive more monitoring than others, which should increase the standard errors and decrease the statistical significance of the estimated coefficients.

Across regions, we estimate significant avoided deforestation in the African subsample, but no robust effects in Asia and South America. The coefficient for Africa implies an 18% decrease in the probability of deforestation relative to the 2011–2016 levels (0.04/0.22) (Fig. 4). As the average deforestation in these cells is 1.86%, the estimated avoided deforestation per year is equal to 495.27 $km^2$. If we apply a carbon density of 6,000 $MtCO_2\,km^{-2}$ (the low end of the valuation for forests in this region[10]), and if we value the amount of avoided carbon emissions resulting from alert subscriptions at the social cost of carbon (US\$50 $t^{-1}$)[11], we calculate the alert system's value at US\$149 million. With a carbon density of 28,100 $MtCO_2\,km^{-2}$ (the estimated value of the dense forests in our sample[1]), we estimate a value of US\$696 million. Both numbers exceed the costs of developing and maintaining the system. Furthermore, this number represents a lower bound, as impacts of using GLAD are larger with a 1 yr lagged specification (Supplementary Tables B14–B16), which suggests that when subscribers have more experience with the alerts, their impact on deforestation is greater. Although some cells had subscriptions for multiple years, these total benefits are only calculated for the last year (when all subscriptions were in place) to avoid overstating the impacts.

Finally, by way of identifying a mechanism through which deforestation might be controlled, we examine whether the effect of subscriptions varies with local policy efforts. For this analysis, we include interactions between subscriptions and indicators for whether or not a grid cell is more than 50% contained by a concession or protected area. Information on concessions is not available for all countries, so these estimates are for the subsample of countries that have both concession and protected areas data (Methods). Marginal effects are presented in Fig. 5 and are extracted from Supplementary Table B2. The individual interactions are negative and statistically significant—subscriptions have a stronger deterrent effect in both protected areas and in forest concessions.



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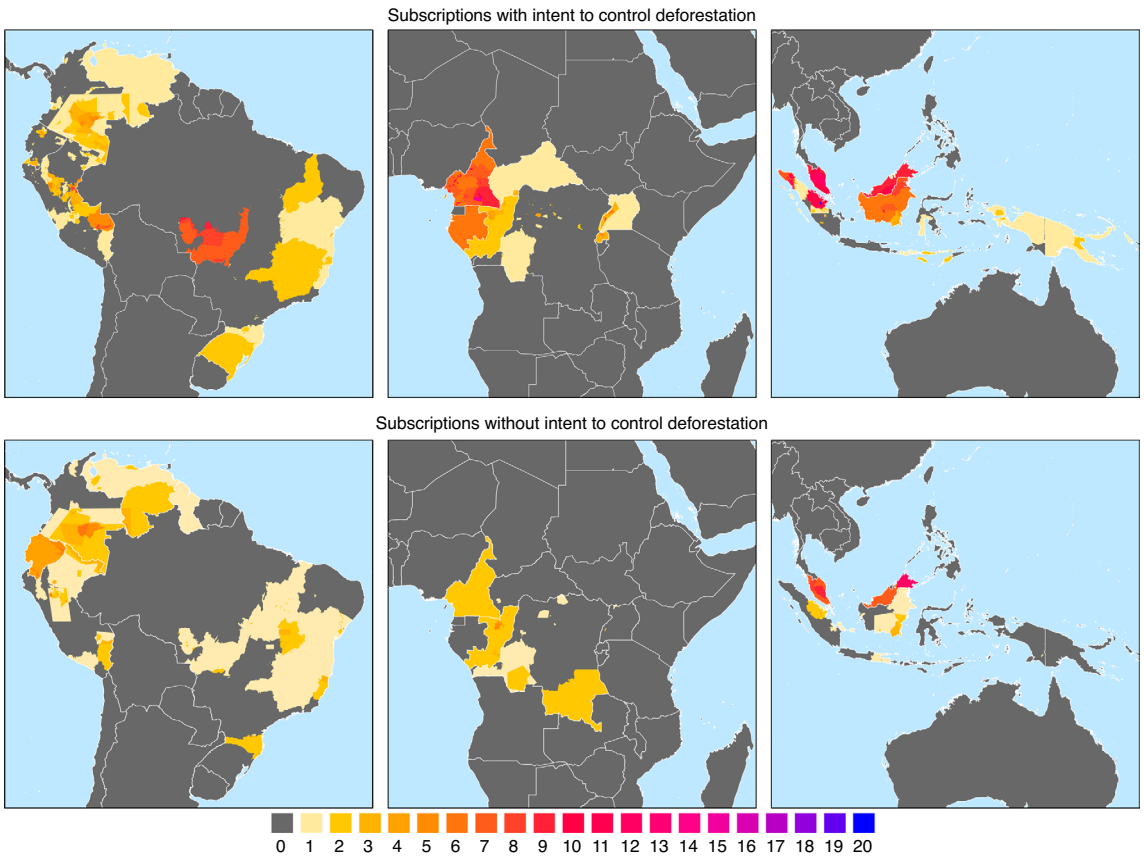



**Fig. 3 | Spatial distribution and number of subscriptions.** Subscriptions with (top) and without (bottom) intent to control deforestation are shown. Darker colours indicate greater numbers of subscriptions.

**Table 1 | Summary statistics**

| | Early | | Late | | Never | | Normalized difference |
|---|---|---|---|---|---|---|---|
| | Mean | s.d. | Mean | s.d. | Mean | s.d. | (early versus late) |
| Outcomes | | | | | | | |
| Deforestation over 2011 to 2016 (0/1) | 0.23 | 0.42 | 0.18 | 0.38 | 0.23 | 0.42 | 0.09 |
| Per cent deforestation over 2011 to 2016 | 0.78 | 4.52 | 0.42 | 2.79 | 0.55 | 2.77 | 0.07 |
| Winsorized per cent deforestation over 2011 to 2016 | 0.30 | 0.81 | 0.21 | 0.67 | 0.30 | 0.81 | 0.08 |
| Controls | | | | | | | |
| Average precipitation per day (mm) | 0.22 | 0.20 | 0.20 | 0.08 | 0.18 | 0.08 | 0.09 |
| Average temperature | 25.85 | 2.71 | 26.35 | 3.05 | 26.01 | 2.57 | −0.12 |
| Protected areas (%) | 12.27 | 32.45 | 21.88 | 41.04 | 13.49 | 33.80 | −0.18 |
| Distance to nearest port within country (km) | 289.64 | 357.25 | 371.06 | 280.97 | 616.58 | 598.17 | −0.18 |
| Distance to nearest road (km) | 7.34 | 11.31 | 20.89 | 25.49 | 10.35 | 15.80 | −0.49 |
| Distance to nearest urban centre with >100,000 inhabitants (km) | 141.52 | 97.05 | 234.90 | 170.47 | 139.54 | 95.04 | −0.48 |
| Number of grid cells | 205,457 | | 194,203 | | 262,844 | | 399,660 |

Early and late subscription groups are based on whether the grid cell had at least one subscription before or by April of 2017 (that is, when 50% of the sample was covered by a subscription; Supplementary Fig. B2). Normalized differences are a scale-free measure of the difference in distributions between samples, which has advantage that it is directly interpretable in terms of how much average standard deviation is the mean from one sample to the mean of the other sample[17].

These mechanisms are particularly relevant for the African subsample—more of the subscribed forests in Africa are in protected areas and concessions (57% versus less than 33% in the other two regions). Separate estimates for these interactions by continent show robust results only for African countries (Supplementary Tables B19–B21). There is also a consistently statistically significant



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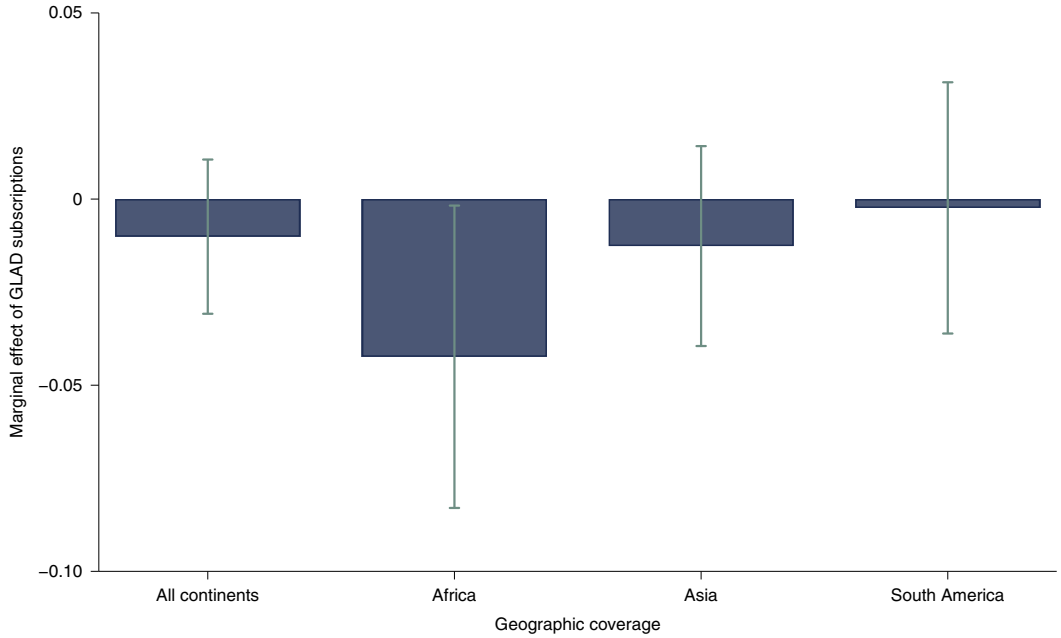



**Fig. 4 | Impact of GLAD subscriptions on deforestation.** The point estimates (shaded bars) and 95% confidence intervals are shown for the impact of GLAD subscriptions with intent to control deforestation on the probability of deforestation on average for all continents as well as for Africa, Asia and South America. Each bar shows the impact from a different estimation: the first is from column 6 in panel A of Supplementary Table B1 and the other three are from column 6 in panel B of the same table, showing the total effect (that is, the linear combination of the baseline and interaction effects).

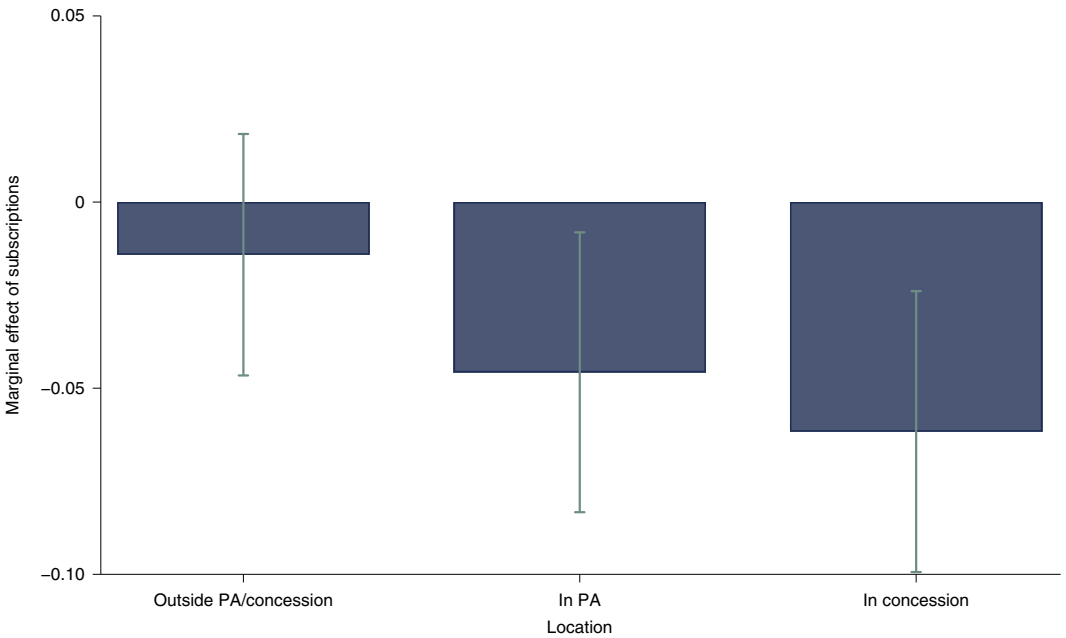



**Fig. 5 | Estimates of GLAD subscriptions in concessions and protected areas on deforestation.** The point estimates (shaded bars) and 95% confidence intervals are shown for the impact of GLAD subscriptions with intent to control deforestation on the probability of deforestation. Each bar shows a marginal effect: the first is from subscriptions in areas without protected areas (PA) or concessions, the second is in protected areas and the third in concessions. The estimates come from column 6 of Supplementary Table B2.

negative impact of concessions with subscriptions in the Asian subsample, although not in protected areas (Supplementary Fig. B3). This effect is not visible in the average impact in Asia because concessions cover a small part of the estimation sample (14%). As our strategy does not account for the endogenous placement of protected areas and concessions, we present these results as intriguing correlations. However, these correlations confirm the anecdotal evidence on alerts: that they are used for enforcing protected area policy and for controlling illegal deforestation within forest concessions.

## Benefits and limitations of use

Our Analysis presents strong evidence that freely available forest-change detection data can support decreased deforestation. The use of GLAD alerts through online subscriptions with intent



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to control deforestation led to an 18% decrease in the probability of forest loss in the African countries included in our sample. The estimated carbon benefits of the avoided deforestation from GLAD subscriptions—from US$149 million US$696 million—is a low-end estimate. It is likely that real benefits are much higher, given that impacts seem to be increasing over time, and that there are a number of substantial co-benefits alongside avoided carbon emissions, including biodiversity and watershed protection, that are not valued in the calculation.

Our empirical approach supports a causal interpretation. Using the timing of availability or subscription to the alert system helps alleviate an important source of bias, as all observed units are eventually affected by the intervention. We control for confounding land characteristics, temporal shocks and regional trends. Although we cannot definitively lay out the causal chain that drives this result, there are a number of reasons why the effects of the alert system were large and statistically significant for only the African countries in our sample. First, countries in Africa had minimal or no broad-scale deforestation monitoring technology before GLAD. Because of this, the introduction of freely available alerts constituted a new source of information to support policy interventions. Indeed, there is evidence that Cameroon's government relies on GLAD for deforestation monitoring[12], and that efforts to stem deforestation in Africa have recently been boosted through regional agreements (https://afr100.org/content/home).

In contrast to Africa, in Asia there is some evidence that the private sector has already made large investments in monitoring of deforestation and fires[13,14]. In this situation the free alert system may have provided additional information for only some areas of forest, as alerts were already available in part of the region. As our identification of effects relies on a break in the deforestation trend when alerts became available or subscriptions activated, there would be no detectable effect where monitoring was already taking place via existing systems. We do observe a correlation between subscriptions in concessions and decreases in deforestation in Asia, but these forests do not cover a large enough area to drive a region-wide impact.

In Latin America, the effects may be muddied both by existing monitoring systems and by political unrest. Regarding existing systems, in 2017, the Peruvian government stopped using GLAD and created their own early warning system[15]. This means that GLAD subscriptions within Peru are not necessarily representative of the areas where the government is taking more action. Colombia and Venezuela, which constitute a substantial portion of the South American sample, were facing political challenges when subscriptions became available in November 2017. Finally, South American countries in our sample, except Peru, have also had access to GLAD for a shorter time period than Africa and Asia. It is likely to take time for the implementation of deforestation policy to effectively use the alert system, as evidenced by our analysis of lag effects.

The alerts that we analyse here are the highest resolution, most accessible and highest frequency system ever created. They are not, however, the first. During the early 2000s, the first near-real-time deforestation monitoring system used coarse-resolution data ($500 \times 500$ m)[16]. These systems were accessible to only a few countries and required specific technical skills to manipulate the data. Until recently, data-driven forest monitoring depended, at best, on yearly forest cover maps[6]. Yearly wall-to-wall maps were relatively scarce until the release of the first worldwide annual deforestation maps[9], which greatly decreased the burden of data access. Still, annual maps are insufficient for stopping deforestation in action, and those available require GIS skills to use effectively. Although GFW has hosted a few lower-resolution and less-accessible alert systems, including FORMA (Forest Monitoring For Action) and Terra-I, the GLAD system represents a substantial improvement in scale, coverage and accessibility (Supplement C).

The alert system has an important limitation: a loss event can only be detected if there are no clouds above it when satellites pass over. If an area of forest is cleared and there are clouds overhead when the satellite images the area, it will not be detected. As images can only be obtained every 8 days, a month or more may pass before an image without clouds blocking the clearing can be obtained, particularly during rainy seasons or in regions with persistent cloud cover. This means that there can be a time lag between the clearing event and the actual alert publication, so interventions to stop further deforestation may also be delayed. This affects their utility, but it does not affect our impact estimates as we do not directly include the alerts in our estimation. The fact that we detect an impact suggests that even this imperfect measure provides additional important information to control deforestation. Furthermore, we do not think that variation in alert lags across continents drives the variation in our results—forests in Africa, Asia and South America have similar cloud-free image availability (an average of 76% months per year) during our study period. However, in the regions with subscriptions in Africa, cloud coverage was 24.7% lower than in South America and 15.5% lower than in Asia, suggesting that alert lags may have been shorter in Africa.

This Analysis contributes to the ongoing discussion about how to address climate change through forest conservation policies in developing countries and via zero-deforestation supply-chain initiatives. It is particularly relevant because a substantial number of developing countries do not have the resources to produce their own deforestation alert system. The globally consistent platform analysed here can also facilitate worldwide investments in climate change interventions.

Important unanswered questions remain about how the benefits of GLAD alerts will evolve in the long term, the most effective ways to encourage use and the technical limits of such large-scale monitoring. Nonetheless, we conclude that the GLAD system of freely available near-real-time forest monitoring provides an immensely valuable scaffold for the implementation of national forest policies, and subsequently to the world that benefits from the global public goods provided by standing forests.

## Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41558-020-00956-w.

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## Methods

**Sample.** We use a stratified random sample of 1 km$^2$ grid cells drawn from across countries with proportional allocation within states. To ensure a representative sample, countries are divided into two groups: those with more than 75,000 forested grid cells and those with fewer than 75,000 forested grid cells. For the first group, we randomly selected 10% of all grid cells. For the second group, we first determined the minimum random sample percentage according to Lohr[18] and then proportionally sample this percentage of grid cells across states within countries. The sample was limited to grid cells that had a minimum of 50% forest cover in 2010 and at least 25 ha of forest with canopy cover ≥60%. Brazilian forests in the Amazon biome are excluded from the analyses due to the existing monitoring system there. We study all countries that had access to the alerts before 2018 (Fig. 1a).

The countries with fewer than 75,000 forested grid cells are: Burundi (70), Rwanda (638), Timor-Leste (3,862), Brunei (4,830), Uganda (8,813), Equatorial Guinea (25,352) and French Guiana (74,855). Using the same formula to sample the grid cells in the countries with more than 75,000 forested grid cells would lead to sampling less than 10%. To ensure representation within countries, sampling for these locations is stratified at the second administrative level. Supplementary Table A11 presents the population of forested grid cells, sample sizes and corresponding sampling percentages. Sampling weights used in empirical estimations correspond to the inverse of the probability that the observation is included due to sampling design.

To control for the lack of precision from GLAD alerts in wetland areas due to changes in water levels, we exclude wetland areas. Specifically, we used the landform map for Indonesia from the University of Maryland[19], the published map of wetlands for the Congo Basin[20] and the Pantanal biome from Brazil. For the countries without a regional map, we used the international CIFOR dataset[21]. The final sample includes 662,504 grid cells.

**Subscriptions.** Subscriptions are classified as with and without intent to control deforestation. We exclude subscriptions for areas greater than 100 Mha (this area corresponds to a little more than the size of a large state (such as Mato Grosso, Brazil) or a small country (Portugal for example) as they are considered too large to be used for monitoring and are most probably associated with users exploring the platform. To create the with-intent group, we exclude all subscriptions associated with the academic sector as well as those subscriptions created by staff from WRI or affiliates (together these constitute the without-intent subscriptions). The end result is a total of 558 subscriptions with intent and 734 without. The number of grid cells ever covered by a subscription with intent is 399,660, whereas those that have ever been covered by a subscription without intent total 298,574.

Of the 558 subscriptions with intent, 302 declared their primary job responsibility using a drop down menu within the alerts system. Supplementary Table B22 details these responses. The majority of these were GIS specialists, with a relatively large number of programme managers, technical staff, land-use planning specialist, reporters, and forest/park managers. Of those that did not answer this question but did declare their location, the largest number was from Indonesia.

The subscriptions layer has time and spatial variation. Each polygon of the layer corresponds to a subscribed area with a date of initiation of the subscription. Although it is the case that some subscribers terminate their subscriptions, to circumvent the potential endogeneity problems with using this variation, we assume that once a subscription has been made, the user remained interested in the subscribed area until the end of our study period. Grid cells are marked as with subscription from the date of the first subscription.

**Annual deforestation and forest cover.** Our main dependent variable is annual deforestation[9] extracted at the grid-cell level for 2011 through 2018. We use this annual measure of deforestation (rather than the sum of alerts for that year) because the alerts are reliant on single observations (making them more susceptible to error) and are designed to be conservative, whereas the annual product maps loss are based on the entire year and can capture the area of loss more accurately[9]. We prefer a binary measure of deforestation that is equal to one if there is a positive amount of deforestation during a given year. The reason for this is that the two continuous outcomes (per cent deforested and a winsorized version of per cent deforested) have substantial outliers, even in the case of the winsorized measure. For example, the per cent deforestation outcome ranges from zero to 100. However, the measure at the 75th percentile is zero, and at the 95th, 3.57. For the winsorized version of this variable, the 75th percentile value is still zero, and the 95th is 3.29. This type of skewness, particularly when combined with measurement error, can result in considerable bias in regressions.

**Other covariates.** We also extract a number of covariates to use as control variables. As weather can influence both the choice to deforest and the measurement of forest cover, data on temperature are extracted from NASA Land surface temperature[22] and precipitation from Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS; ref. [23]). To control for transportation costs (and because deforestation frontiers are known to be more intensely concentrated near urban centres), we calculate three measures of distance. First, the distance to the nearest urban centre with more than 100,000 inhabitants, which can be downloaded from the Joint Research Centre's Global Human Settlement Layer[24]. Second, the distance to the nearest port according to the World Port Index from the National Geospatial-Intelligence Agency[25]. Third, the distance to the nearest road was calculated on the basis of the original road data of OpenStreetMap (available from Geofabrik[26]). Identifiers for the second administrative level are assigned to each grid cell using version 3.6 of GADM (ref. [27]).

We determine whether a grid cell is contained within a protected area using lists of worldwide protected areas provided by the IUCN and UNEP-WCMC (ref. [28]), and available on the GFW website. Data identifying the location of forest concessions come from various country-specific sources and are available on the GFW website. About half of the countries in our sample provided data on concessions in at least one of the following categories: logging, oil palm and wood fibre. Countries are Brunei, Cameroon, Central African Republic, Democratic Republic of the Congo, Equatorial Guinea, Gabon, Indonesia, Malaysia, Papua New Guinea and the Republic of Congo. Finally, for each grid cell, the per cent observed per year is the average per cent per month of the area with forest cover in 2010 with at least one clear observation (excludes areas obscured by cloud or shadow).

**Estimation strategy.** The basic estimation strategy exploits variation in either access to alerts (to estimate the impact of availability) or use of subscriptions. The estimation equation is:

$$d_{i,c,k,y} = \beta \mathrm{GLAD}_{i,c,y} + \psi_{i(c)} + \alpha_{k,y} + \mu' \mathbf{X}_{i,c,k,y} + \epsilon_{i,c,k,y}$$

where $d_{i,c,k,y}$ is a binary variable indicating deforestation in grid cell $i$ in country $c$ of continent $k$ in year $y$. $\mathrm{GLAD}_{i,c,y}$ is equal to the share of months that the country had access to GLAD alerts in year $y$ or, to assess the impact of subscriptions, an indicator equal to one once over 50% of a grid cell fell within a subscribed area for the first time. In the first case, because the treatment variable only changed across countries and time, we first collapse the data to the country-year level (eliminating the subscript $i$ in the equation). In the second, the treatment variable varies within countries and across time, so we run the estimation at the grid-cell level. $\beta$ is the estimated treatment effect. The samples differ across estimates as well. To estimate the impact of availability, we use the sample of cells that had access to GLAD alerts by 2018 and to assess the impact of subscriptions, those with a subscription at any time before 2018.

We also include country (estimation of availability) or grid-cell (estimation of use) fixed effects, $\psi_{i(c)}$, which control for time-invariant characteristics of the country or grid-cell affecting deforestation (such as soil suitability and slope). In both specifications, we also include continent-year fixed effects, $\alpha_{k,y}$, which control for general macroeconomic events occurring for all countries within the same continent and influencing overall deforestation trends (global food demand, world agricultural prices and so on); and a vector of controls $\mathbf{X}_{i,c,k,y}$ that includes: average temperature, cumulative precipitation, interactions of year indicators with the per cent area in protected areas at the baseline and the distances to the nearest port, nearest road and nearest city. The distance/time interactions are included to help eliminate confounding variation resulting from changes in transport or production technologies that would affect the time trends of deforestation in more remote areas differentially. $\mu$ is the vector of estimated parameters. Owing to the small number of clusters in the country-level estimation, we calculated confidence intervals using wild bootstrap clustering (a method that relies on random sampling with replacement to calculate the confidence intervals). Residuals ($\epsilon_{i,c,k,y}$) were clustered at the second administrative level for the subscription estimations. We apply sampling weights in each regression so that point estimates can be considered to represent the population of forest.

**Reporting Summary.** Further information on research design is available in the Nature Research Reporting Summary linked to this article.

## Data availability

Datasets analysed for this study are available from the corresponding author upon reasonable request.

## Code availability

The codes used to generate the figures and tables are available via Zenodo[29].

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## Author contributions

F.M. and J.A.-G designed research, performed econometric analyses and led the writing. K.S. managed the data compilation and contributed to the writing. A.H.P contributed to the database.

## Competing interests

The authors declare no competing interests.

## Additional information

**Supplementary information** is available for this paper at https://doi.org/10.1038/s41558-020-00956-w.

**Correspondence and requests for materials** should be addressed to F.M.

**Peer review information** *Nature Climate Change* thanks Juliano Assunção, Johannes Reiche and Juan Robalino for their contribution to the peer review of this work.

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*Our web collection on statistics for biologists contains articles on many of the points above.*

## Software and code

Policy information about availability of computer code

| | |
|---|---|
| Data collection | All data except the subscriptions and the percent observed (i.e. percent of the grid-cell that is not obscured by cloud or shadow) were downloaded from publicly available websites. They were spatially compiled at the grid-cell level by ArcMap version 10.6 and the package ArcPy. Different projections were used when creating the grid-cells: South America Albers Equal Area Conic, Africa Albers Equal Area Conic, and Asia South Albers Equal Area Conic. For the first analysis (i.e. impact of GLAD availability), data were collapsed down to the country level with the standard "collapse" command in Stata-SE version 15.1. |
| Data analysis | The main analysis was conducted using the standard "xtreg" command in Stata-SE version 15.1 |

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The codes used to generate the figures and tables are available in the Zenodo data repository (https://doi.org/10.5281/zenodo.4054099). All data necessary for replication of the results in this paper are available from the corresponding author on reasonable request. Country and state GADM codes correspond are available for download from GADM version 3.6 at https://gadm.org/data.html. The original Tree Cover 2010 and Deforestation data can be downloaded from https://earthenginepartners.appspot.com/science-2013-global-forest/download_v1.6.html. We excluded grid-cells in wetland areas using the wetland (landform) map for Indonesia from Margono et al. (2014) (https://glad.umd.edu/dataset/primary-forest-cover-loss-indonesia-2000-2012), the map for the Republic of the Congo and the Democratic Republic of the Congo from Bwangoy et al. (2010) (http://dx.doi.org/10.1016/j.rse.2009.08.004), the map from NASA for the Amazon Basin (http://







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dx.doi.org/10.3334/ORNLDAAC/1284), and the Pantanal biome from Brazil. For the countries without a regional map, we used the international CIFOR dataset (https://www.cifor.org/globaladditional -wetlands/). The original precipitation data from CHIRPS can be downloaded from https://www.chc.ucsb.edu/data/chirps. The original temperature data from NASA Land surface temperature can be downloaded from https://neo.sci.gsfc.nasa.gov/view.php?datasetId=MOD_LSTD_M. The original protected areas and concession areas (logging, oil palm, and forest) can be downloaded from www.globalforestwatch.org. The original major urban center, defined as cities above 100,000 people, can be downloaded from JRC's Global Human Settlement (https://ghsl.jrc.ec.europa.eu ). The original port data can be downloaded from World Port Index from the National Geospatial-Intelligence Agency (https://www.arcgis.com/home/item.html?id=b04b76b94059436e93757c301c10026c). The original road data (OpenStreetMap) can be downloaded from Geofabrik (https://www.geofabrik.de/data/download.html). The original subscription layer with information on subscribers is available from World Resources Institute but restrictions apply to the availability of these data. However, the grid-cell level subscription timing are included in the replication files available from the authors upon reasonable request. Original subscription layer with information on subscriptions can be obtained upon reasonable request and with permission of World Resources Institute. The percent observed (i.e. percent of the grid-cell that is not obscured by cloud or shadow) has been created by the University of Maryland and is available in the replication files available from the authors upon reasonable request.

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# Behavioural & social sciences study design

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| | |
|---|---|
| Study description | Quantitative methods were used to analyze the effect of the availability and the use of real-time forest monitoring on deforestation. |
| Research sample | We use a sample of 1 x 1 km grid-cells randomly drawn from forested grid-cells across all countries that began receiving alerts prior to 2018. These countries are: Peru, Republic of the Congo, Brazil (outside of the Amazon biome), Brunei, Malaysia, Indonesia, Papua New Guinea, Timor Leste, Burundi, Cameroon, Central African Republic, Equatorial Guinea, Gabon, Democratic Republic of the Congo, Rwanda, Uganda, Colombia, Ecuador, French Guyana, Guyana, Suriname, and Venezuela. |
| Sampling strategy | For the countries with less than 75,000 forested grid-cells, we follow Lohr (2000). For the countries with more than 75,000 forested grid-cells, we randomly draw 10% of all grid-cells. To ensure representation within country, the random sampling is stratified at the second-administrative level. |
| Data collection | Not applicable. |
| Timing | Study covers the 2011-2018 period. |
| Data exclusions | Brazilian forests in the Amazon biome are excluded from the analyses due to the existing monitoring system there since the mid-2000s. |
| Non-participation | Not applicable. |
| Randomization | Not applicable. |

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